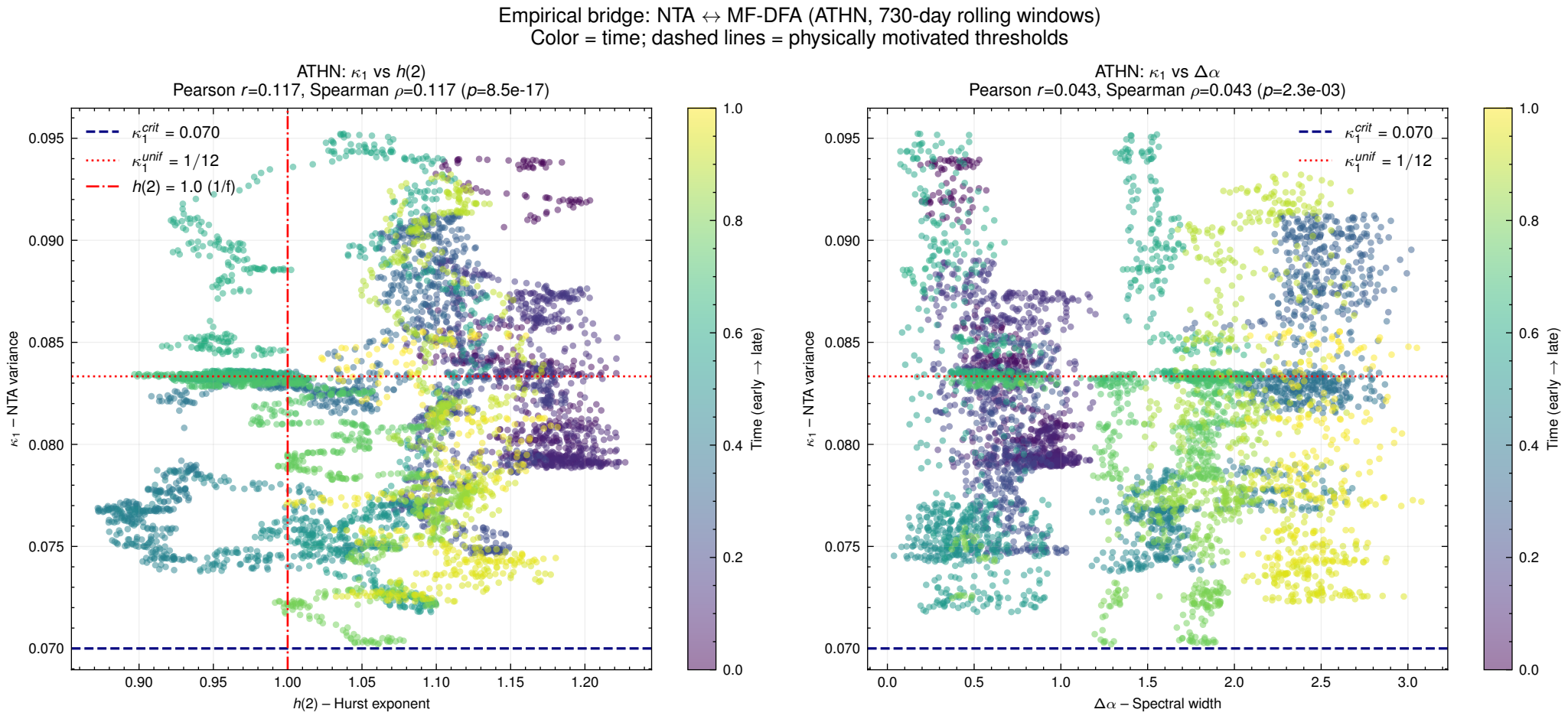
Empirical bridge: NTA ↔ MF-DFA (ATHN, 730-day rolling windows)
Color = time; dashed lines = physically motivated thresholds
ATHN: κ_1 vs $h(2)$
Pearson r=0.117, Spearman ρ=0.117 (p=8.5e-17)
κ_1^{crit} = 0.070
κ_1^{unif} = 1/12
$h(2)$ = 1.0 (1/f)
κ_1 – NTA variance
$h(2)$ – Hurst exponent
Time (early → late)
ATHN: κ_1 vs $\Delta\alpha$
Pearson r=0.043, Spearman ρ=0.043 (p=2.3e-03)
κ_1^{crit} = 0.070
κ_1^{unif} = 1/12
κ_1 – NTA variance
$\Delta\alpha$ – Spectral width
Time (early → late)